Assignment 5 Task 2 Report

A character-level decoder-only Transformer was implemented from scratch on Shakespeare text. The best baseline used four layers, four heads, width 128 and learned positions, achieving validation cross-entropy 2.6520 and perplexity 14.1831.

Data and training setup

Setting	Value
Corpus	6,248,564 characters; vocabulary of 78 characters
Split	First 90% training: 5,623,707; final 10% validation: 624,857
Sequences	128 characters; targets shifted forward by one character
Batches	64 sequences; 686 training and 76 validation batches; drop_last=True
Optimizer	AdamW; learning rate 0.0003; gradient clipping at 1.0
Training	20 epoch maximum; dropout 0.1; initial random seed 42
Baseline grid	$L \in \{2, 4\}$; $h \in \{2, 4\}$; $d_{model} \in \{64, 128\}$; learned or sinusoidal positions
Depth study	$L \in \{2, 4, 6, 8\}$; $h = 4$; $d_{model} = 128$; learned positions

Model and formulas

Character embeddings and positional encodings enter pre layer-normalized blocks with residual connections. Each block contains causal self-attention and a GELU feed-forward network of width 4d_model. A final layer normalization and linear head predict the next character.

$Q = XW_Q$, $K = XW_K$, $V = XW_V$; $d_k = d_{model} / h$

$\text{Attention} = \text{softmax}(QK^T / \sqrt{d_k} + M)V$

$M(i, j) = 0$ when $j \le i$; $M(i, j) = -\infty$ when $j > i$

$\text{Loss} = -(1/BT) \sum_b \sum_t \log p(x_{t+1} \mid x_{\le t})$; $\text{PPL} = \exp(\text{Loss})$

$PE(pos, 2i) = \sin(pos / 10000^{(2i/d_{model})})$

$PE(pos, 2i+1) = \cos(pos / 10000^{(2i/d_{model})})$

The causal mask prevents access to future characters. Early stopping checks $|\text{average loss}(k) - \text{average loss}(k-1)| < 10^{-4}$. The saved runs completed 20 epochs; reported scores are final epoch scores rather than best epoch checkpoints.

Baseline validation results

All 16 baseline configurations are sorted by final validation loss. L denotes layers, h denotes heads and d denotes embedding width. Lower validation loss and perplexity are better.

L	h	d	Positions	Train loss	Val loss	PPL
4	4	128	learned	1.1363	2.6520	14.1831
4	2	128	sinusoidal	1.1353	2.7856	16.2100
2	2	128	sinusoidal	1.2182	2.8795	17.8052
2	2	64	sinusoidal	1.3844	2.8985	18.1474
4	2	128	learned	1.1436	2.9680	19.4538
4	4	128	sinusoidal	1.1254	2.9852	19.7899
2	4	128	sinusoidal	1.2189	3.0881	21.9352
2	2	64	learned	1.4281	3.1719	23.8525
4	4	64	learned	1.3432	3.2122	24.8333
2	2	128	learned	1.2588	3.2659	26.2042
4	2	64	learned	1.3459	3.3251	27.8006
4	2	64	sinusoidal	1.3041	3.3780	29.3118
2	4	128	learned	1.2478	3.3911	29.6982
4	4	64	sinusoidal	1.2931	3.4366	31.0818
2	4	64	sinusoidal	1.4160	3.5085	33.3978
2	4	64	learned	1.4773	4.0610	58.0305

Brief comparison

Increasing the head count did not consistently improve validation performance. At L = 4 and d = 128, learned positions performed best with four heads, while sinusoidal positions performed better with two heads. Positional encoding performance therefore depended on the model configuration.

The lowest training loss was not the best validation result. The gap between training and validation loss suggests limited generalization on the final contiguous portion of the corpus; the split may also contain different plays or text structure.

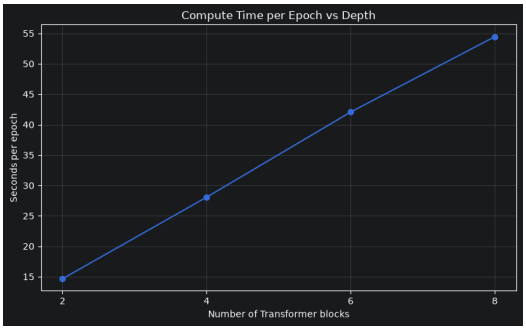


Figure 1. Measured time per epoch versus depth in the separate depth scaling runs. Timings include training and validation, averaged over completed epochs.

Baseline loss curves

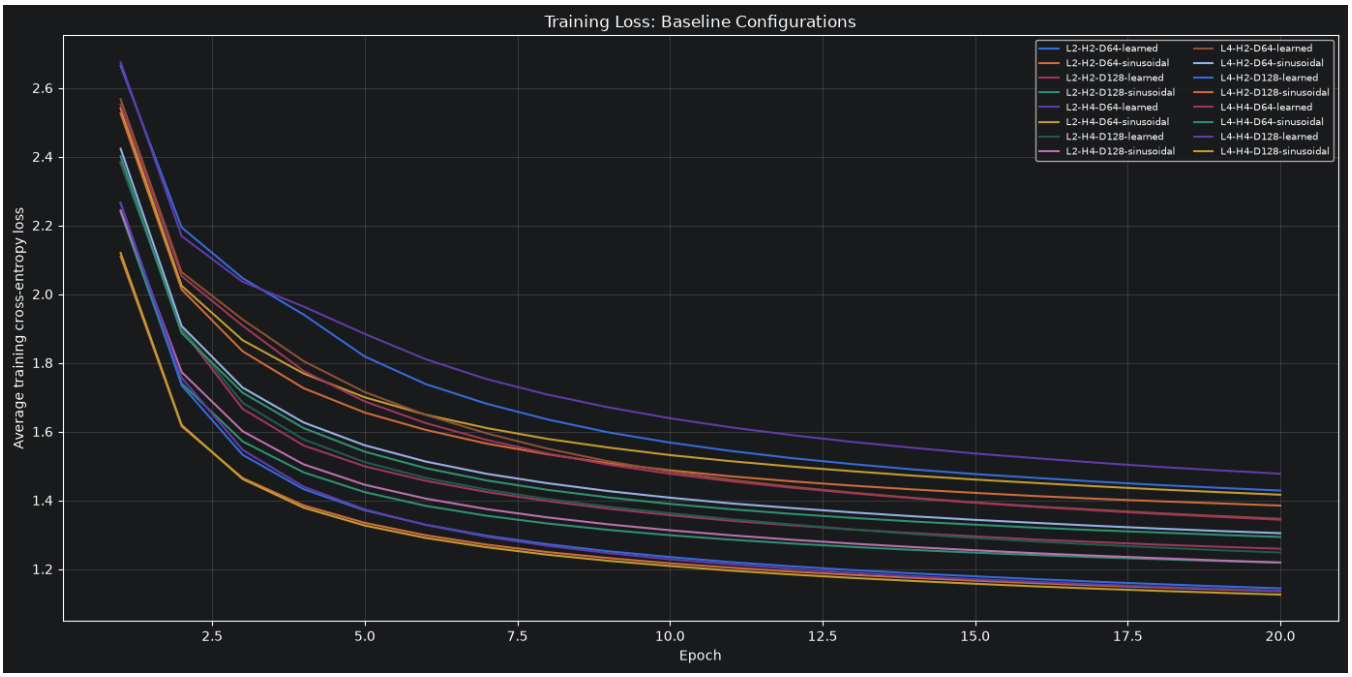


Figure 2. Average training cross-entropy across all 16 baseline configurations.

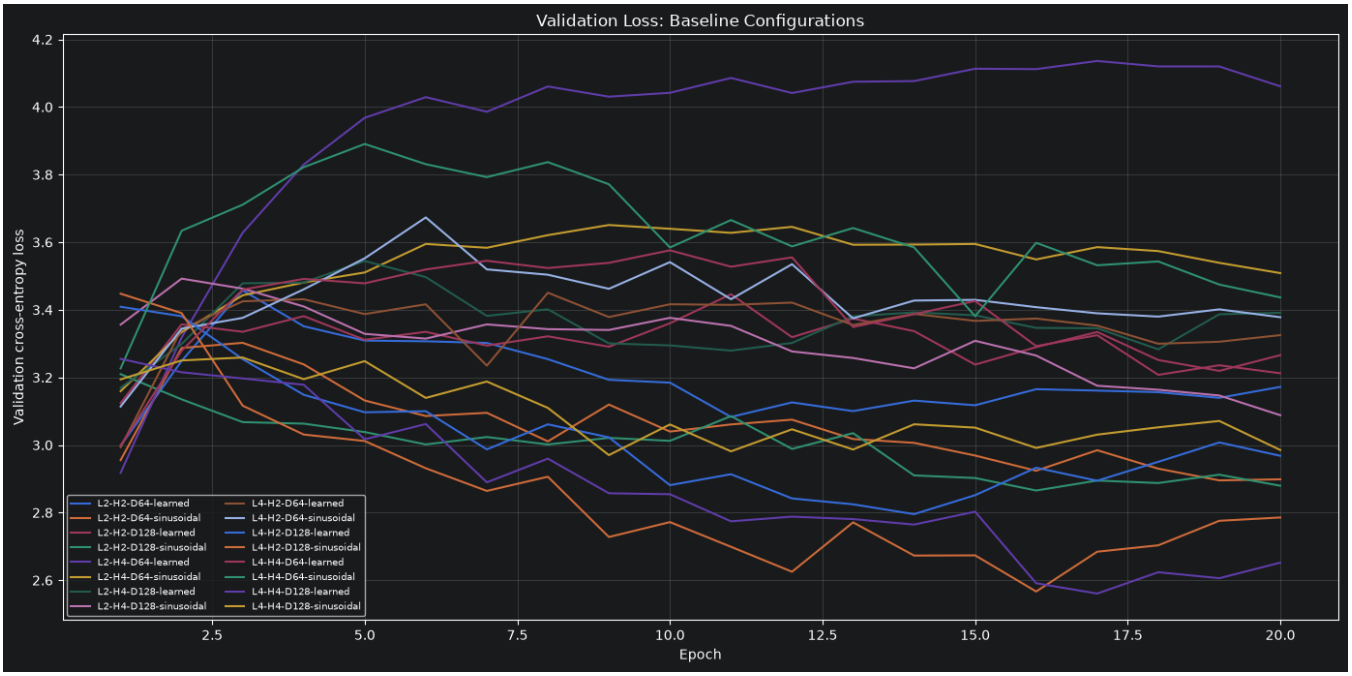


Figure 3. Validation cross-entropy across the same configurations.

Training loss decreased steadily, while validation curves fluctuated and remained substantially higher. Additional training did not uniformly improve validation performance.

Depth scaling results

Layers	Train loss	Val loss	PPL	Seconds per epoch
2	1.2415	2.9678	19.4483	14.6570
4	1.1330	2.8210	16.7931	28.0688
6	1.0943	3.1334	22.9515	42.0721
8	1.0669	3.0056	20.1981	54.4561

These are separately trained depth study runs, so the two and four layer scores differ from the baseline grid. Four layers gave the lowest final validation loss in this study. Eight layers reduced training loss further but took about 3.72 times the two layer time per epoch.

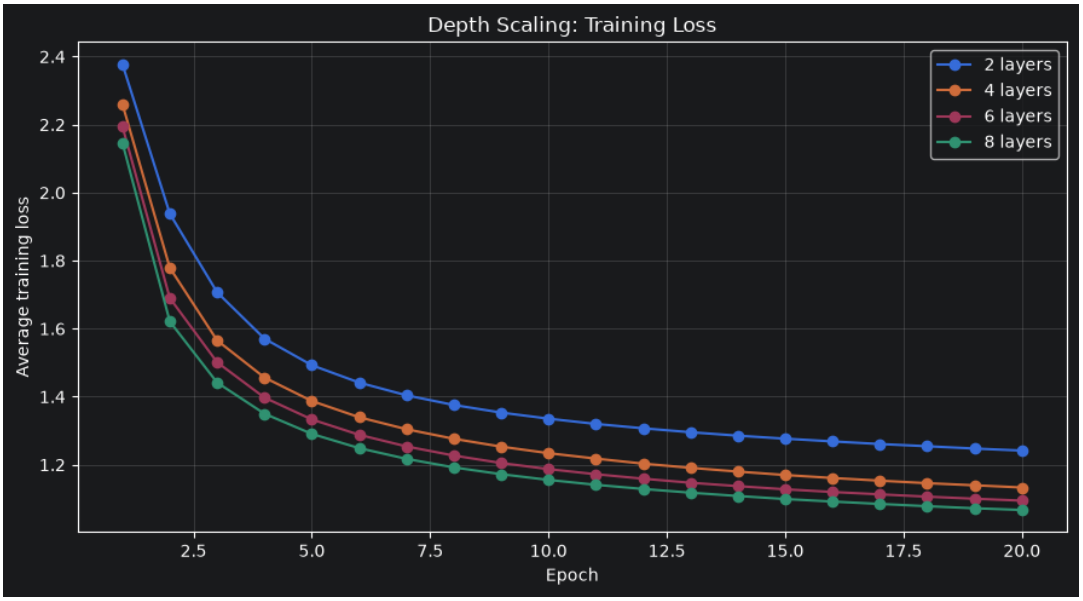


Figure 4. Training loss for the depth scaling study.

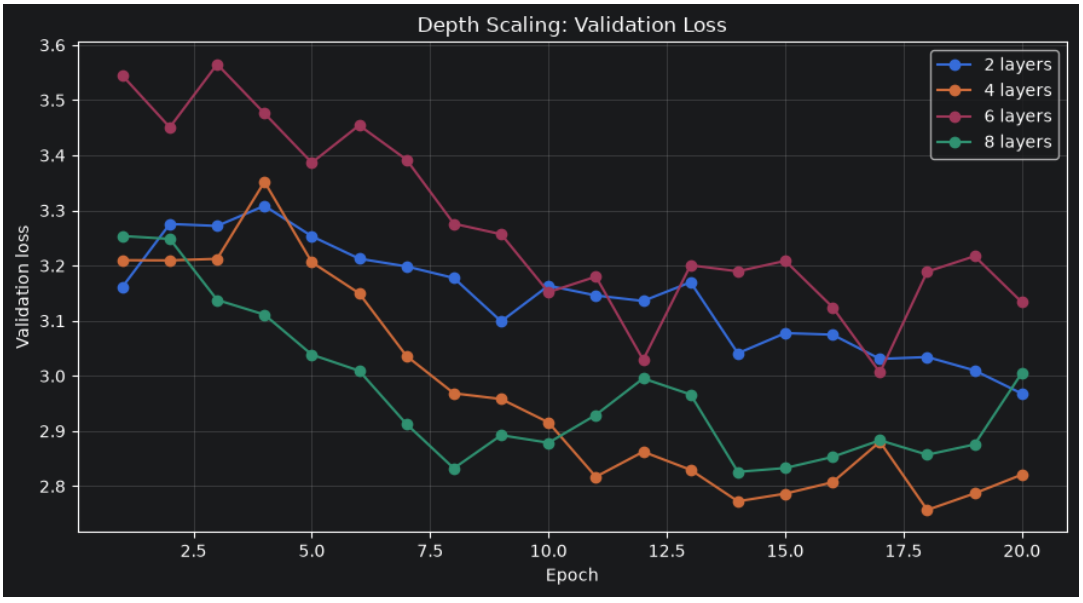


Figure 5. Validation loss for the depth scaling study. Greater depth did not give a consistent validation benefit.

Generated text examples

The selected baseline model generated 250 new characters per prompt. The five saved examples below cover five prompts, greedy decoding and all three temperatures. Outputs are reproduced as saved, including incomplete words and grammatical errors.

Greedy: next character = argmax(logits). Sampling: $p_i = \exp(z_i / \tau) / \Sigma_j \exp(z_j / \tau)$. Lower τ concentrates probability; higher τ increases diversity.

To be, or not to be greedy temperature: N/A

```
To be, or not to be so much a strange of the sea,"
"And the strange of the stranger of the state,"
"And the strange of the stranger of the state,"
"And the stranger of the state of the state,"
"And the stranger of the state of the state,"
"And the stranger of the state
```

KING: My lord sampling temperature: 0.5

```
KING: My lord. Why, sir, the counteries of the"
"heart of the land, not have been the manner, and stranger"
"The great of the life, and the lovers of my hand,"
"Then must not to the langer of the world,"
"And fare the field with others and infection"
"That thou a
```

ROMEO: sampling temperature: 0.8

```
ROMEO: EXETRUS, and with all the forward, but fork"
"Each great Shall traitor of Bargumuris, and this"
"Fall what patricus are scried again."
"Take his privatience that holds with garments"
"Greem and crown to come to him life,"
"The less of whom! all the
```

What is the cause sampling temperature: 1.2

```
What is the cause,"
"Let sit the bosom wipper"
"Valarity since to us, let me strong'd, Crence,"
"Beside.--I, 'He lief hither fum back she."
"Knows so."
"Ay, sick, ah, mastern as I am Ether."
"Play to my haunty:"
"To make us dignias your importain,"
"Hod Earl sfys, wh
```

The night is dark sampling temperature: 0.8

```
The night is dark in the prithee for a paper"
"That reason to the good love, instantly paintainty."
"And be intered of mers of your life, if you"
"And you like a thing brief, poor make woman?"
"Away you do to you see the proud of your desiress,"
"Hold elder but some
```

Greedy decoding repeated phrases such as “the stranger of the state”. Sampling at $\tau = 0.5$ was more conservative; $\tau = 0.8$ and 1.2 produced more varied but less grammatical text. The model captured some dramatic style without reliable long-range coherence.

Causal attention heatmaps

The following maps show the first block of the selected baseline model for the first 20 characters of a validation sequence. Rows are query characters and columns are key characters.

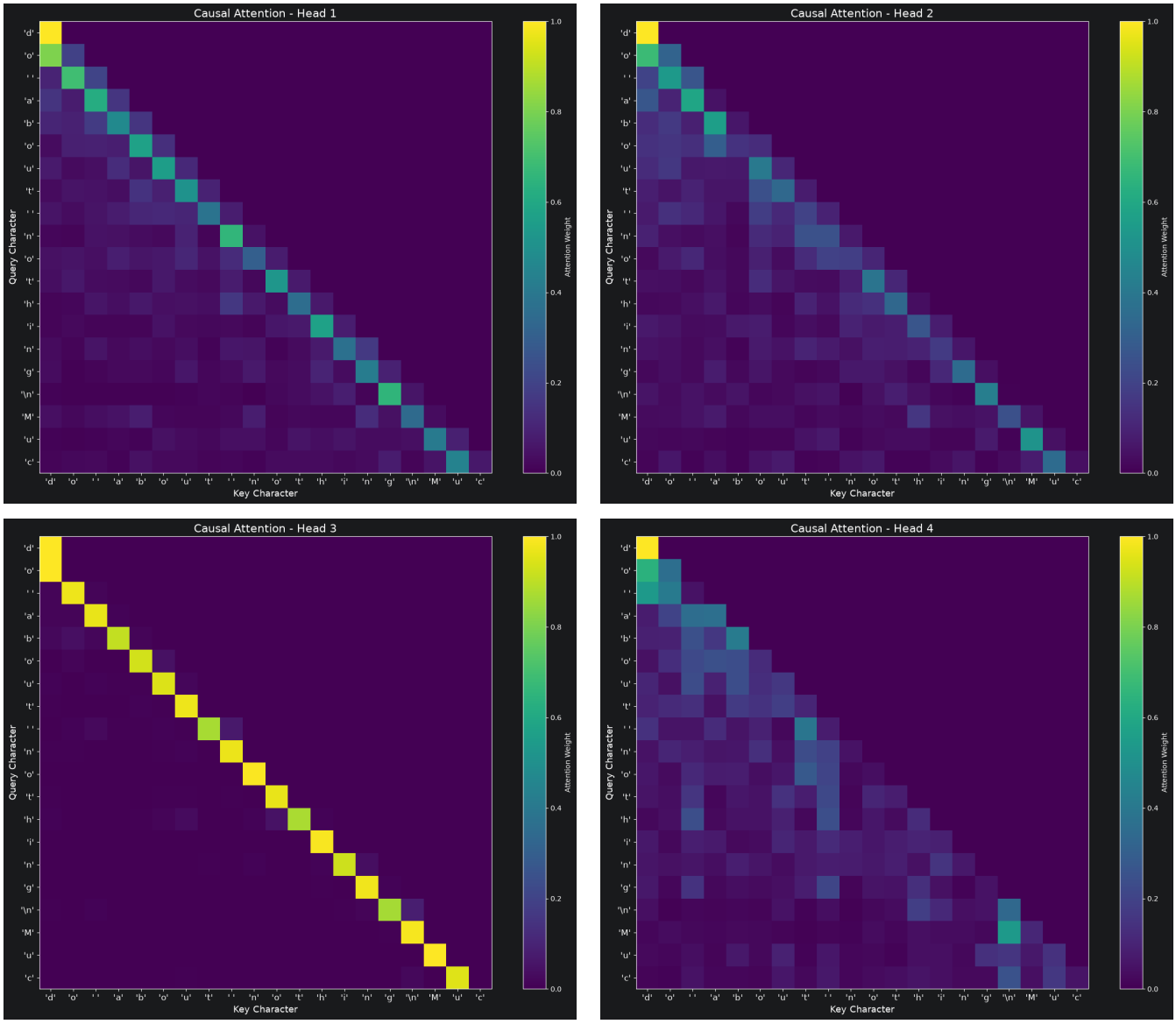


Figure 6. All four attention heads. Entries above the diagonal are zero because future keys are masked; the heads distribute attention differently over the allowed prefix.

Findings

The best baseline reached perplexity 14.1831. Four layers offered the strongest final validation result in both the baseline grid and the separate depth study. Deeper models increased compute cost without a consistent improvement in validation loss.

The notebook selected models using final validation loss. It did not restore the lowest validation loss epoch, and seeds were not reset for each configuration. The comparisons describe the saved single runs; differences cannot be attributed to one hyperparameter alone with statistical confidence.